I asked prompted NotebookLM to give me a quiz with the following prompt: “Please give a quiz that asks about SSMs as in the lectures, discussions and homework. The problems should be reasonably difficult, not just a basic overview of what was discussed. Please also come up with a numerical examples that will test understanding of how an SSM works (so hand computing some details of SSMs).”

The quiz itself is reasonably good, but with several mistakes when making computational questions and the LLM’s answers to the problem, as shown here:

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In a standard State-Space Model (SSM) defined by $h_{t+1} = Ah_t + Bx_t$ and $y_t = Ch_t$, let $A = \begin{pmatrix} 0.5 & 1 \\ 0 & 0.2 \end{pmatrix}$, $B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, and $C = \begin{pmatrix} 2 & 1 \end{pmatrix}$. Assuming the initial state $h_0 = 0$, what is the second term of the convolution kernel, K_1, which corresponds to the coefficient of x_{k-1} in the expansion of y_k?

A. 3.2

B. 3.0

× Not quite

This result might be obtained if the matrix C was $\begin{pmatrix} 2 & 1 \end{pmatrix}$ instead of $\begin{pmatrix} 2 & 2 \end{pmatrix}$.

C. 4.0

✓ **Right answer**

The first term of the kernel (after the D term, which is 0) is $K_1 = CB$. Computing this gives $CB = \begin{pmatrix} 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = (2)(1) + (1)(1) = 3$. Oh wait, the source defines the kernel as $(D, CB, CAB, \ldots)$. So the second term is $K_1 = CB$. Let me re-read the question. Ah, it asks for the *second* term of the kernel, which is indeed $K_1 = CB$. So $CB = [2, 1] * [1; 1] = 2 * 1 + 1 * 1 = 3$. Let me re-check the math. Hmm. Let's re-read the lecture notes. `yk = ... + CAB uk-2 + CB uk-1 + Duk`. The kernel is `K = (D, CB, CAB, ...)`. So $K_0 = D, K_1 = CB, K_2 = CAB$. The question asks for K_1. My calculation of $CB = 3$ is correct. Let me re-evaluate the options. Maybe there's a different convention? Let me look at the homework `hw08_solution.pdf`. It says `K = (D, CB, CAB, CA2B, ... )`. My calculation is correct. Let me check the provided options again. Maybe I'm supposed to do CAB? $AB = \begin{pmatrix} 0.5 & 1 \\ 0 & 0.2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1.5 \\ 0.2 \end{pmatrix}$. Then $CAB = \begin{pmatrix} 2 & 1 \end{pmatrix} \begin{pmatrix} 1.5 \\ 0.2 \end{pmatrix} = 3 + 0.2 = 3.2$. This is K_2. Let me re-calculate CB. I must have made a simple error. $C = [2, 1]$, $B = [1; 1]$. $CB = 2 * 1 + 1 * 1 = 3$. None of the options is 3. This is strange. Let me re-read the problem statement in my generation. Ah, I made a mistake in creating the problem. Let me fix the problem and options. Let's set $C = \begin{pmatrix} 2 & 2 \end{pmatrix}$. Then $CB = \begin{pmatrix} 2 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 2 * 1 + 2 * 1 = 4$. This works. $CAB = C(AB) = \begin{pmatrix} 2 & 2 \end{pmatrix} \begin{pmatrix} 1.5 \\ 0.2 \end{pmatrix} = 3 + 0.4 = 3.4$. Okay, the question is now correct. The rationale for the correct answer should be: The second term of the convolution kernel is given by $K_1 = CB$. The calculation is $\begin{pmatrix} 2 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 4$.

However, it is really good at creating conceptional questions based on the given sources.

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The S4 model parameterizes the eigenvalues of its diagonal state matrix A as $\lambda = \exp(-\mathrm{ReLU}(\lambda_R) + j\lambda_I)$. What is the primary motivation for this specific formulation?

A. To ensure the magnitudes of the eigenvalues are always non-negative and less than or equal to 1, which aids gradient-based optimization and prevents the system's state from exploding.

✓ **That's right!**

The $-\mathrm{ReLU}(\lambda_R)$ term ensures the real part of the exponent is non-positive, constraining $|\lambda| \leq 1$. This prevents exploding states while allowing for stable, long-range dependency modeling on the unit circle.

B. To constrain the magnitude of all eigenvalues to be strictly less than 1, ensuring the system's history always decays rapidly.

C. To force the eigenvalues to lie on the imaginary axis, which is required for modeling periodic signals with perfect reconstruction.

D. To ensure all eigenvalues are real numbers, which simplifies computation and avoids complex arithmetic.